

VAPO Hits 60.4 on AIME 2024: +10 vs DAPO

Value-model-based RL for advanced long-chain reasoning

Headline result: VAPO reaches 60.4 (avg@32) on AIME 2024 using Qwen2.5-32B without SFT data.

Beats prior baselines: DAPO (50) and DeepSeek-R1-Zero-Qwen-32B (47), i.e., 10+ point gain.

Reaches SOTA within ~5,000 training steps and reports zero training crashes across multiple runs.

Core claim: value-model-based RL can outperform value-free methods when engineered for long CoT.

Why Value-Model-Based RL Matters for Long CoT

Token-level credit assignment: traces step-wise actions to final return.

Lower variance than pure Monte Carlo trajectory-level assignment.

Potentially better sample efficiency via generalized value estimation.

Value-free methods (e.g., GRPO, DAPO) assign final reward uniformly over tokens.

That simplification can hide local reasoning errors in long trajectories.

VAPo revisits value models with targeted mechanisms for long-CoT failure modes.

Three Core Challenges in Value-Model Training

1) Value Model Bias

Long trajectories + bootstrapping amplify estimation bias.

2) Heterogeneous Lengths

Different response lengths need different bias-variance tradeoffs.

3) Reward Sparsity

Only final <eos> token receives reward signal.

VAPO Framework: Techniques Mapped to Challenges



Deep Dive: Length-Adaptive GAE for Heterogeneous Responses

Novel mechanism: adjust GAE λ by response length instead of fixed λ globally.

Short responses use lower λ (lower variance); long responses use higher λ (lower bias).

Targets the unique long-CoT regime where response lengths vary dramatically.

Ablation signal: removing Length-adaptive GAE reduces AIME24 from 60 to 45 (−15).

Interpretation: dynamic bias-variance control is essential for stable token-level credit assignment.

Main Results: VAPO Establishes New SOTA on AIME 2024

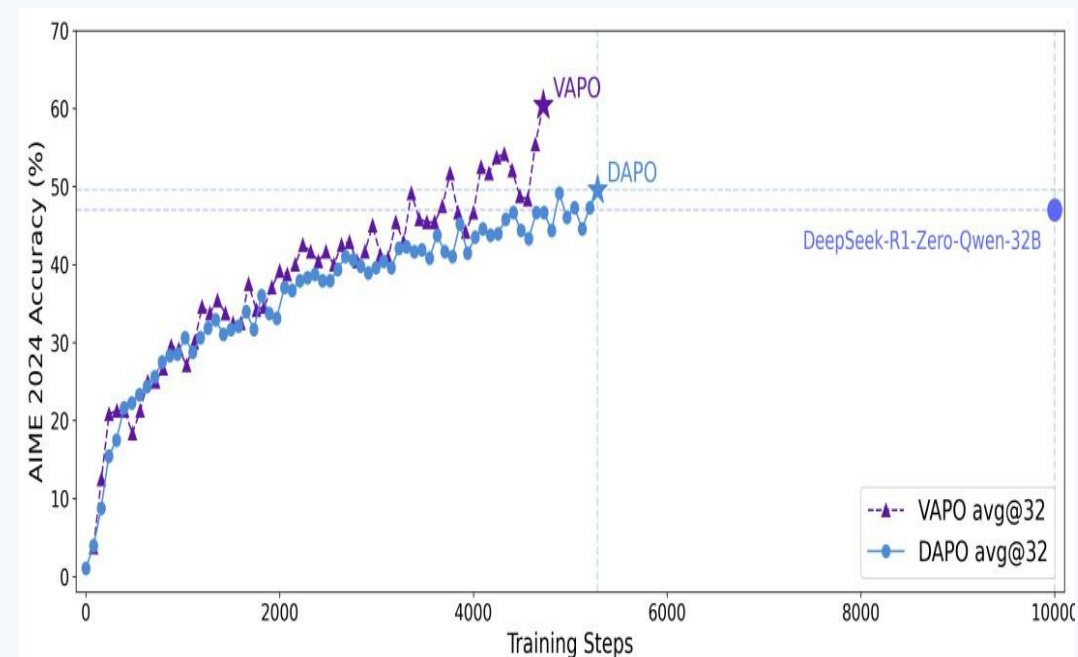
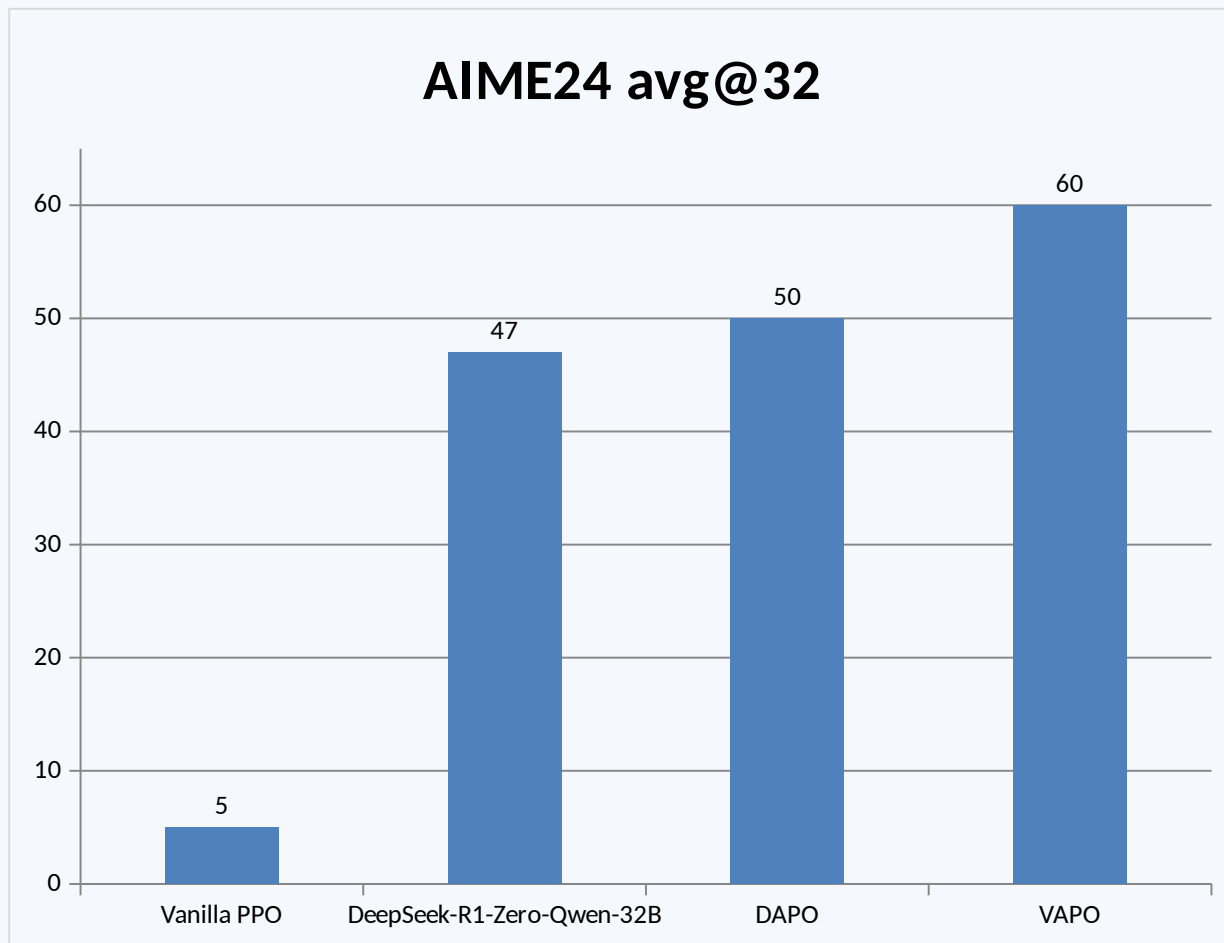
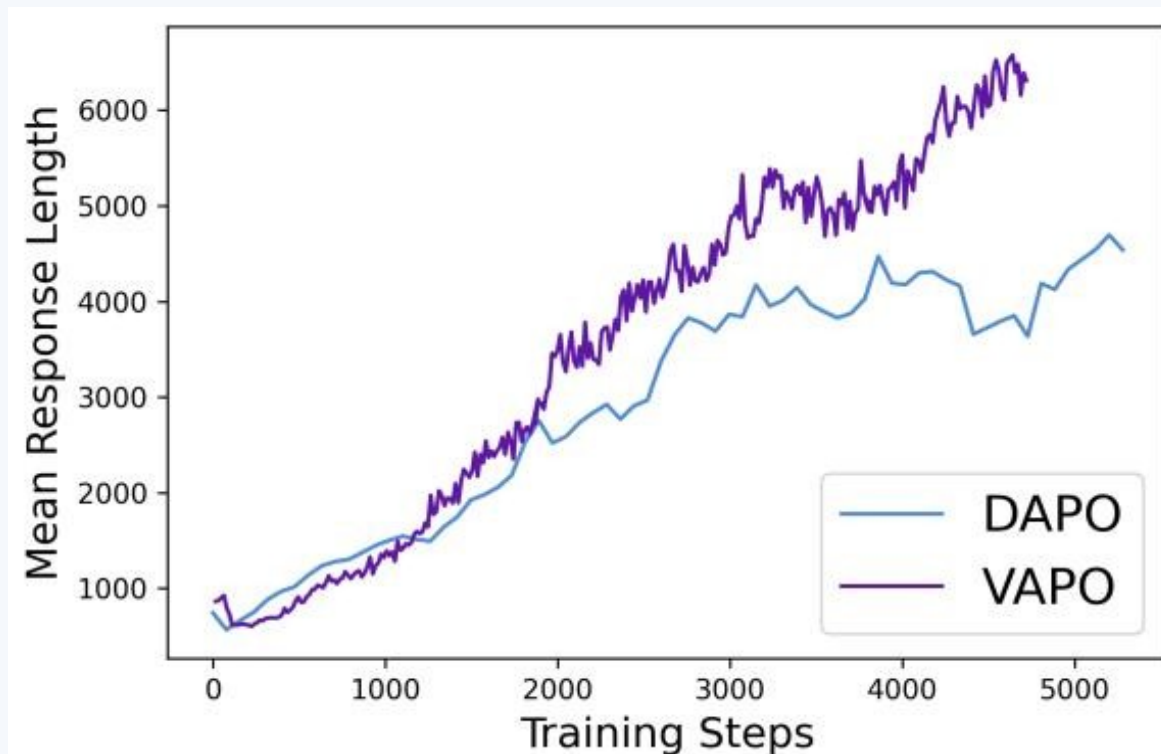
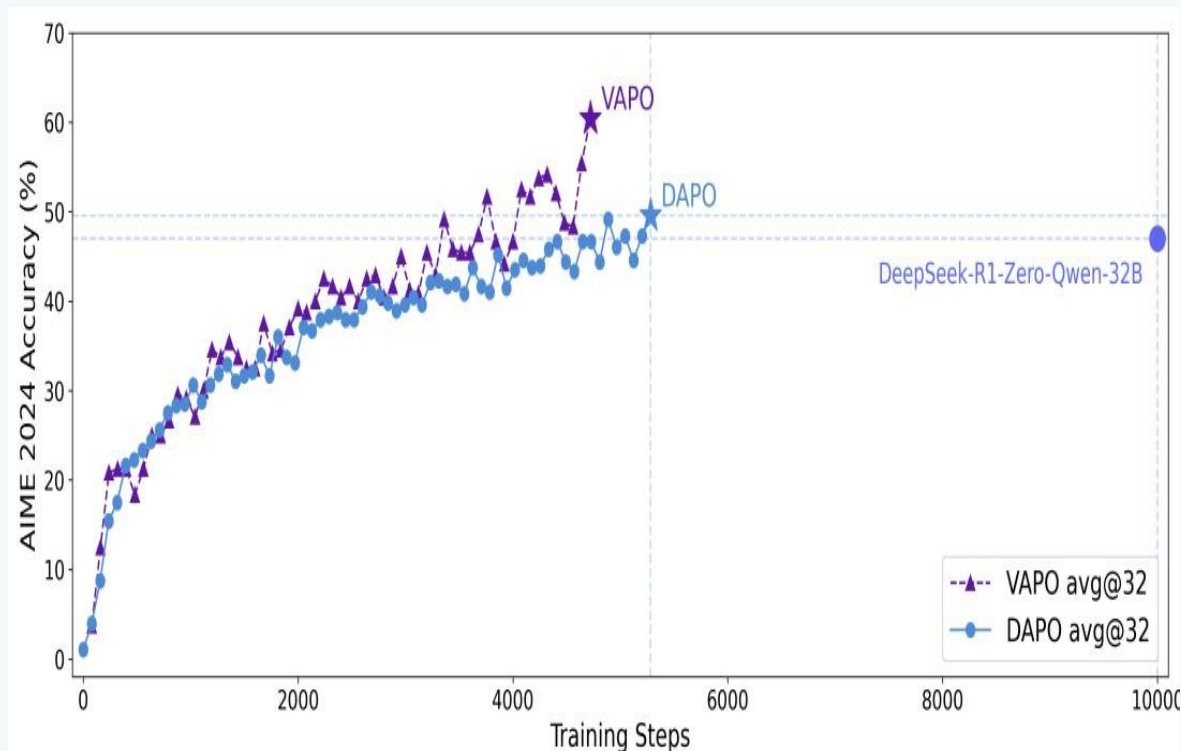


Figure 2: Training curve comparison (VAPO vs DAPO).
VAPO reaches ~60 around step 4,500; DAPO plateaus near 50 around step 5,500.

Training Efficiency: Faster Convergence, Longer Reasoning Chains



VAPO reaches SOTA in ~5K steps and generates longer outputs (~6,000 tokens vs ~4,500 for DAPO), indicating stronger long-form reasoning behavior.

Ablation: Every Component Matters (AIME24 avg@32)

Variant	Score	Delta vs Full
VAPO (full)	60	—
w/o Value-Pretraining	11	-49
w/o Decoupled-GAE	33	-27
w/o Length-adaptive GAE	45	-15
w/o Clip-Higher	46	-14
w/o Token-level Loss	53	-7
w/o Positive Example LM Loss	54	-6
w/o Group-Sampling	55	-5

Largest single effect: removing Value-Pretraining drops performance from 60 to 11 (-49 points).

Key Takeaways and Implications for Reasoning RL

Value-model-based RL can surpass value-free baselines when bias, length heterogeneity, and reward sparsity are jointly addressed.

Length-adaptive GAE is a critical mechanism for long-CoT settings with variable response lengths.

Value-Pretraining is the highest-impact component in VAPO (-49 points when removed).

VAPO combines higher accuracy (60.4), strong efficiency (~5K steps), and robust stability (no reported crashes across runs).

Implication: future reasoning model development should re-invest in value modeling with targeted engineering for long-horizon credit assignment.